

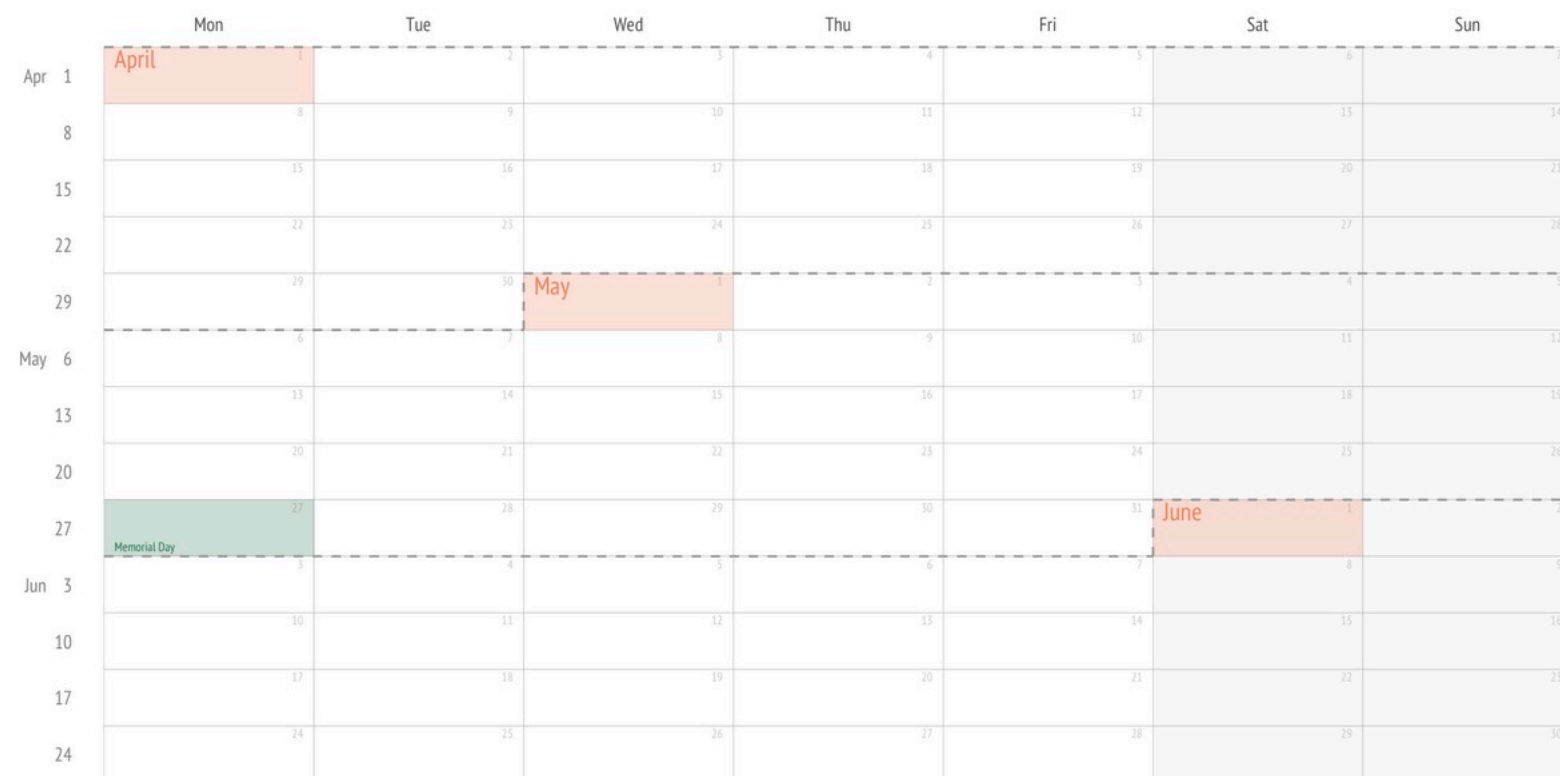
Reusing 'ggplot2' code: how to design better plot helper functions

Combined Melbourne R User and Statistical Society of
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Convenience vs. flexibility

```
1 library(ggweekly)
2 ggweek_planner(
3   start_day = "2019-04-01",
4   end_day = "2019-06-30",
5 )
```



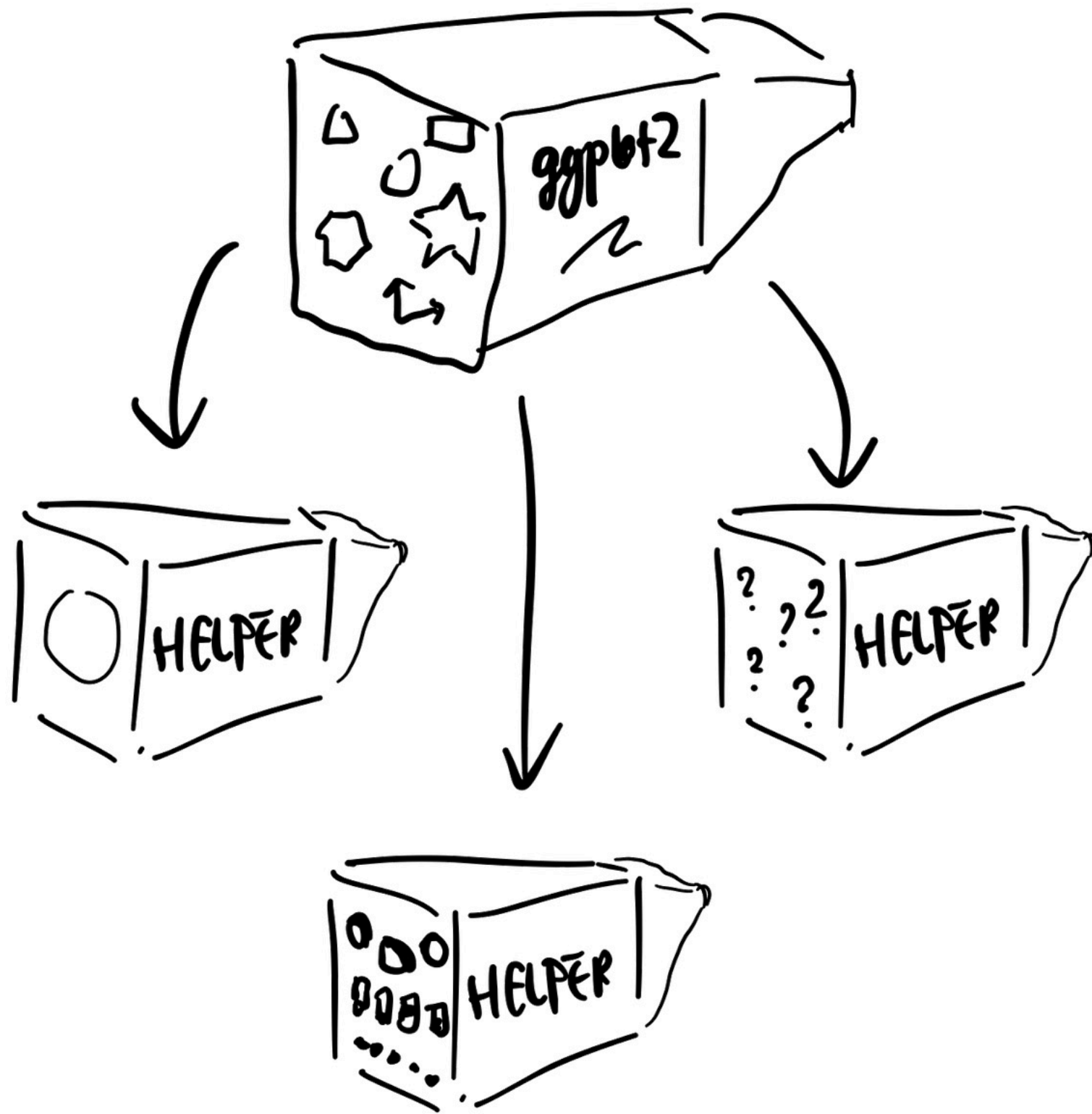
What about customisation?

```
1 ggweekly::ggweek_planner(
2   start_day = lubridate::today(),
3   end_day = start_day +
4     lubridate::weeks(8) - lubridate::days(1)
5   highlight_days = NULL,
6   week_start = c("isoweek", "epiweek"),
7   week_start_label = c("month day", "week", "
8   show_day_numbers = TRUE,
9   show_month_start_day = TRUE,
10  show_month_boundaries = TRUE,
11  highlight_text_size = 2,
12  month_text_size = 4,
13  day_number_text_size = 2,
14  month_color = "#f78154",
15  day_number_color = "grey80",
16  weekend_fill = "#f8f8f8",
17  holidays = ggweekly::us_federal_holidays,
18  font_base_family = "PT Sans",
19  font_label_family = "PT Sans Narrow",
20  font_label_text = NULL
21 )
```

Example from github.com/gadenbuie/ggweekly

Beware false savings!

Sometimes
“saved” lines of code
turn into
**confusing lists of plot
specific arguments**



Design goals

- Quick ‘autoplotting’
- Minimise plot-specific arguments
- Retain ggplot syntax & error messages
- Integrate well with other ggplot2 components

Convenient autoplotting

ggtilecal 0.1.0

ggtilecal

The goal of `ggtilecal` is to easily produce calendar layouts using ggplot2 tile and text geoms, while retaining some customisation.

Installation

You can install the development version of `ggtilecal` like so:

```
# install.packages("remotes")
remotes::install_github("cynthiahqy/ggtilecal")
```

Examples

Empty Calendar

```
1 library(ggtilecal)
2 dates <- make_empty_month_days(
3   c("2024-01-05", "2024-06-30"))
4
5 gg_facet_wrap_months(
6   dates,
7   unit_date)
```

Jan							Feb							Mar						
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	2	3	4	5	6	7				1	2	3	4					1	2	3
8	9	10	11	12	13	14	5	6	7	8	9	10	11	4	5	6	7	8	9	10
15	16	17	18	19	20	21	12	13	14	15	16	17	18	11	12	13	14	15	16	17
22	23	24	25	26	27	28	19	20	21	22	23	24	25	18	19	20	21	22	23	24
29	30	31					26	27	28	29				25	26	27	28	29	30	31

Apr							May							Jun						
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	2	3	4	5	6	7				1	2	3	4						1	2
8	9	10	11	12	13	14	6	7	8	9	10	11	12	3	4	5	6	7	8	9
15	16	17	18	19	20	21	13	14	15	16	17	18	19	10	11	12	13	14	15	16
22	23	24	25	26	27	28	20	21	22	23	24	25	26	17	18	19	20	21	22	23
29	30						27	28	29	30	31			24	25	26	27	28	29	30

Supporting customisation

ggtilecal 0.1.0

☰

Make Monthly Calendar Facets

Source: [R/gg_facet_wrap_months.R](#)

Generates calendar with monthly facets by:

- Padding event list with any missing days via `fill_missing_units()`
- Calculating variables for calendar layout via `calc_calendar_vars()`
- Returning a ggplot object as per Details.

Usage

```
gg_facet_wrap_months(  
  .events_long,  
  date_col.
```

Explain:

1. Data preparation steps
2. How the ggplot2 recipe works
3. How to modify the plot

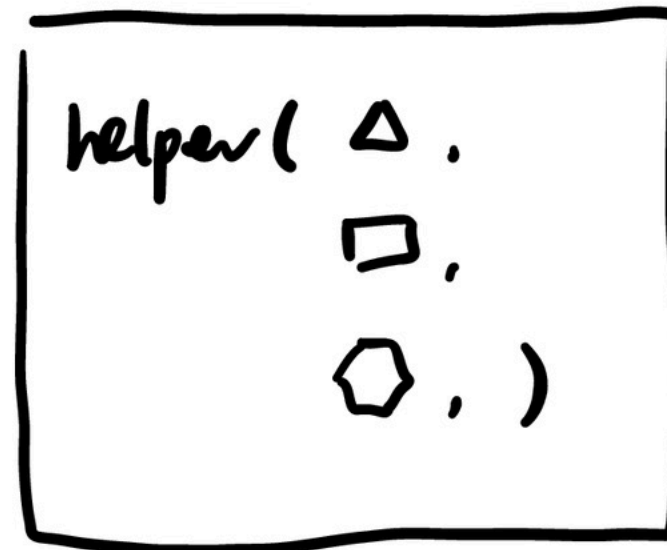
Design principles for plot helpers

1. Expose ggplot syntax & recipe

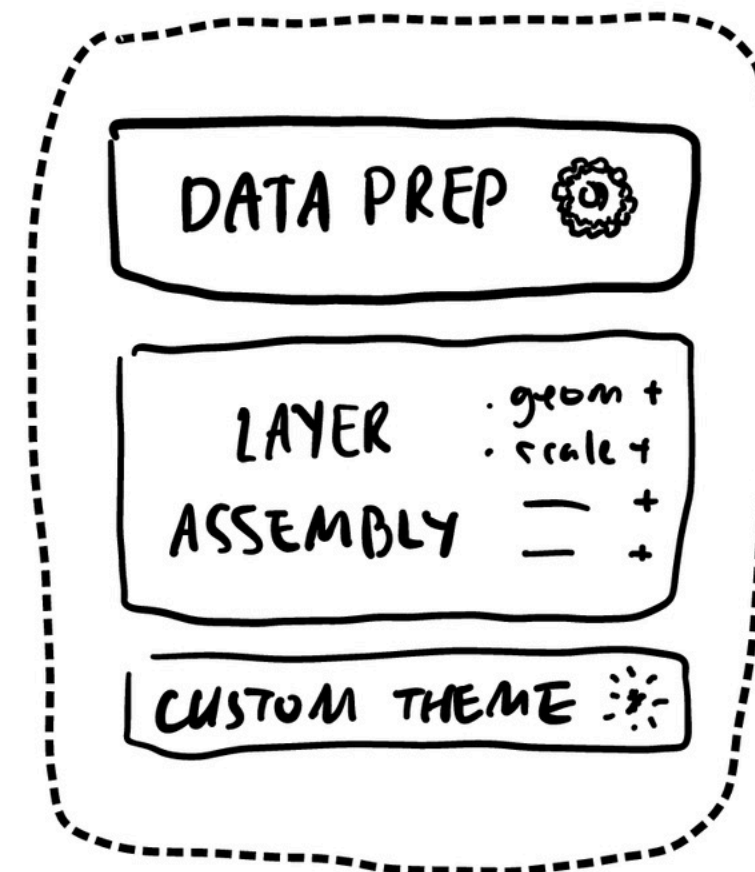
2. Separate functionality

3. Document grammatical customisation

GRAMMATICAL
INPUTS with
autoplot DEFAULTS



MODULAR
INTERNALS



SUPPORT
grammatical
CUSTOMISATION



But remember rules are made to be broken!